

Unstructured data cabling. Data cabling in a traditional data centre runs under the floor right along with the power cables. The cables are long patch cables with the entire under-floor space acting like a cabling trough.

In the past, there were many types of cables under the floor: Cabling for Ethernet Category 3/5/6, single- and multi-mode fibre, twinax, coax, various SCSI types, ESCON, serial, various phone system trunk cables and station cables in 25-400 pair configurations. Today, many cable types have become obsolete and are being replaced with Category 5/6 copper and fibre-optic cables. This greatly reduces the sheer volume and weight of cables in a data centre.

Computer systems rarely last more than five years before they need to be upgraded or replaced. As various new systems are brought into a data centre, little or no thought is given to the cabling. As such, extra-long patch cables are ordered because it is always better to have a cable that is too long than too short, right? Well, yes to a point, but using 10m long fibre cables to connect a storage area network (SAN) switch to 40 blade servers in the rack next door makes for a large pile of fibre that needs to be stored at some place, and that some place is almost always under the raised floor or on top of the rack because it will not fit in the rack and looks messy.

When time comes to move a system, removing the tangled cables from under the floor without impacting another critical system can be time-consuming and result in unscheduled down time. As cables are pulled, they catch on the coils and create knots. This endangers production systems, especially if the power cords are part of the tangled mess.

An option is to just abandon the cable under the floor and put more in for the new systems. Over time, the space below the floor becomes congested with cables.

Fig. 1 shows power distribution in a traditional data centre. There are redundant power feeds (110V or 220V)

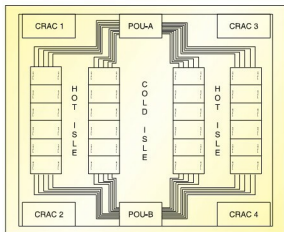


Fig. 1: Power distribution in a traditional data centre (courtesy: Marty Poniatowski)

to each rack. It requires 24 dual-pole circuit breakers in each power distribution unit (48 breaker positions) to provide redundant 220V power to each rack.

Typically, raised-floor data centres use the space under the floor also as a cool-air distribution system for the room. Large computer room air-conditioners (CRAC) are positioned around the room with air intakes at the top and the cold air supply being forced down under the floor. Proper airflow requires ample space so that air can flow under the raised floor. Large piles of data cables, piping, power cables, fire suppression systems and sensors all take up space under the floor and restrict airflow.

High-density racks. With the adoption of hot-aisle/cold-aisle rack layouts, many existing data centres received a life extension of more than five years, but the additional density of computing platforms was just getting started. With the introduction of the blade server, even the hot-aisle/cold-aisle design could not keep up with the heat loads and power requirements.

Traditional racks had migrated from 1 kW per rack with the traditional rack layout to 3-5 kW for a hot-aisle/cold-aisle design. With blade chassis being loaded into racks with up to six blade chassis per rack, the heat load potential for a rack jumps to a staggering 24-30 kW per rack. The apparent answer to this problem appears to be a

design that decreases the density back to a manageable 3-5 kW per rack.

It turns out that the cost per square foot for the real estate and construction costs to build a larger data centre, at least in a metropolitan area, are so great that these negate the benefits of traditional cooling. Typically, data-centre raised floors are 45.7 cm (18 inches) high or less. To provide adequate cooling in higher-density environments, the

floor height rapidly becomes a limiting factor. For example, to support a 24kW rack environment (400 watts/square foot), you need over 1.2 metres (4 feet) of clear height under the raised floor.

Using traditional CRAC systems, most data centres with 45.7-91.4cm high raised floors can expect to support a heat load of up to 6 kW per rack with an average of 3-5 kW. Moving beyond this range requires a different approach to both cooling and power.

The two main factors in cooling data centres are temperature and airflow. CRAC units provide cooling and airflow as well as a humidity control system to add or remove moisture from the air. It is important to have enough moisture in the air to prevent static electricity but not so much as to create the opportunity for condensation.

Large CRAC units operate most efficiently when the airflow entering through the intake on the unit is the hottest air in the room. The system is capable of cooling only the hot air that reaches it. All other air in the room is cooled by mixing the hot air with the cold air from the CRAC. If the hot air is not efficiently returned to the CRAC but allowed to mix with the cool air before reaching the intake, the CRAC will not operate at peak performance, causing wear and tear of the unit.

The lower the temperature of a computer room, the lower the safe humidity level, because lower tem-